



IRFANA ASLAM

Nationality: Pakistani **Date of birth:** 01/06/1998 **Gender:** Female

Phone number: (+92) 3299106379 **Email address:** irfanaaslam69@gmail.com

LinkedIn: [IrfanaAslam](#)

GitHub: [IrfanaAslam](#)

Home: Johar Town, LAHORE (Pakistan)

ABOUT ME

AI-focused Computer Scientist and Bioinformatician with strong research and practical experience in **medical image analysis, deep learning, machine learning, and computational biology**. Proven ability to design, implement, and evaluate end-to-end AI systems using Python, PyTorch/TensorFlow, and computer vision techniques. Published researcher with interdisciplinary expertise spanning **AI for healthcare, bioinformatics, and data-driven modeling**. Actively developing reproducible, research-oriented codebases on GitHub. Highly motivated to pursue doctoral research in **medical imaging, foundation models, generative AI, and multimodal learning**.

RESEARCH INTERESTS

Medical Image Analysis & Computer Vision

Foundation Models for Healthcare AI

Generative AI & Diffusion Models

Vision-Language & Multimodal Learning

Robustness, Generalization, and Data-Efficient Learning

EDUCATION AND TRAINING

Master of Science: Computer Science

COMSATS University of Information Technology, Punjab, Pakistan [09/2021 – 06/2023]

City: Sahiwal | **Country:** Pakistan | **Final grade:** CGPA: 3.25 / 4.00 | **Thesis:** Fabric Design Defect Detection using Deep Learning

- Designed CNN-based deep learning models for image defect detection
- Focused on feature extraction, model optimization, and generalization
- Technologies: Python, TensorFlow, OpenCV, Deep Learning

Bachelor of Science: Bioinformatics

COMSATS University of Information Technology, Punjab, Pakistan [09/2017 – 08/2021]

City: Sahiwal | **Country:** Pakistan | **Final grade:** 2.73/4 | **Thesis:** Next-Generation Multi-Epitope Vaccine Design

- Applied computational biology and immunoinformatics techniques
- Focused on in silico vaccine design and molecular analysis

PUBLICATIONS

[2023]
Design and Assessment of a Novel In Silico Approach for Developing a Next-Generation Multi-Epitope Universal Vaccine Targeting Coronaviruses Design and Assessment of a Novel In Silico Approach for Developing a Next-Generation Multi-Epitope Universal Vaccine Targeting Coronaviruses *Microorganisms*, 11(9):

PROJECTS

[20/09/2021 – 26/06/2023]

Fabric Design Defect Detection using Deep Learning

- Developed a **CNN-based deep learning model** to detect defects in fabric designs for quality control.
- Improved **defect classification accuracy**, reducing manual inspection efforts in textile manufacturing.
- **Technologies:** Python, TensorFlow, OpenCV, CNN, Deep Learning.

[10/09/2020 – 10/06/2021]

Next-Generation Multi-Epitope Vaccine Design (Computational Biology)

- Designed a **novel multi-epitope vaccine candidate** for coronaviruses using in silico approaches.
- Published research in **high-impact scientific journals**.
- **Technologies:** Python, BioPython, Molecular Docking, Immunoinformatics.

[01/01/2019 – 10/05/2020]

Machine Learning Model for Disease Prediction

- Created a **predictive model** for diagnosing diseases based on patient symptoms and test results.
- Achieved **high accuracy** in disease classification using real-world medical datasets.
- **Technologies:** Scikit-learn, Pandas, NumPy, Machine Learning.

[15/10/2022 – 15/06/2023]

Automated Resume Screening System using NLP

- Developed an **AI-driven resume screening system** to rank candidates based on job fit.
- Automated HR recruitment, reducing manual effort by **60%**.
- **Technologies:** Python, NLP, Spacy, TensorFlow.

[21/09/2021 – 10/03/2022]

Real-Time Sign Language Recognition

- Built a **deep learning model** to recognize and translate sign language gestures into text.
- Increased accessibility for the hearing-impaired community.
- **Technologies:** Python, OpenCV, TensorFlow, CNN.

[10/10/2023 – 31/12/2024]

AI-Based Surveillance System for Threat Detection

- Implemented an **AI-driven security monitoring system** for real-time **threat detection**.
- Integrated **YOLOv5** for object detection, enhancing security operations.
- **Technologies:** OpenCV, YOLOv5, Deep Learning, Python, Flask.

Competitor Pricing Scraper

- Automated web scraping system for competitor price monitoring
- **Tools:** Python, Web Scraping

Sales Data Analytics Dashboard

- Business intelligence dashboard for sales analysis
- **Tools:** Python, Pandas, Visualization

Loan Default Prediction System

- Predictive analytics model for financial risk assessment
- **Tools:** Python, Scikit-learn

COVID-19 Data Dashboard

- Interactive dashboard for COVID-19 statistics and trends
- **Tools:** Python, Data Visualization

Smart Energy Intelligence Platform

- Data-driven energy monitoring and analytics system
- **Tools:** Python, Data Analytics

Smart Energy Predictor

- Built ML-based energy consumption prediction system
- **Tools:** Python, Machine Learning

WORK EXPERIENCE

 **Punjab Safe City Authorities** – Lahore, Pakistan

Police Communication Officer

[01/11/2024 – Current]

Monitored surveillance systems and live CCTV feeds for public safety. Managed emergency calls and coordinated timely responses with law enforcement. Maintained incident logs and ensured accurate documentation. Utilized GIS mapping for precise location data during incidents. Ensured compliance with protocols while handling sensitive information. Delivered prompt customer service and resolved citizen queries. Conducted real-time monitoring and reported suspicious activities.

 **Scout Talent Inc** – Remote, Canada

Data Analyst

[20/01/2023 – 30/12/2024]

Analyzed large datasets using Python libraries such as Pandas, NumPy, and Matplotlib to derive actionable insights and support decision-making. Developed and implemented data cleaning and preprocessing pipelines to ensure data quality and accuracy. Created and maintained interactive dashboards and visualizations to effectively communicate findings to stakeholders. Applied statistical analysis and machine learning models using Python to identify trends, patterns, and predictions. Automated data collection and reporting processes to streamline workflows and improve efficiency.

 **Ambleside Consulting Inc** – Remote, United States

Website Developer

[05/01/2020 – 30/10/2024]

Designed and developed responsive websites, ensuring high performance and compatibility across various devices and browsers. Implemented front-end technologies (HTML, CSS, JavaScript) to create interactive and visually

appealing user interfaces. Developed and maintained server-side logic and databases using PHP and MySQL. Customized content management systems (CMS) like WordPress for easy content management and updates. Applied SEO best practices to improve website visibility and search engine rankings.

 **Humber College** – Remote, United States

Content Writer

[01/2019 – 30/12/2023]

Applied strong writing, editing, and proofreading abilities to produce high-quality content. Conducted in-depth research on various topics to ensure accuracy and clarity. Tailored content and style according to client briefs and requirements.

SKILLS

AI & ML

Machine Learning / Deep Learning / Scikit-learn / CNNs / Computer Vision / TensorFlow/Keras

Web & Application Development

Django / React / Webflow / Flask / HTML / CSS

Data Analysis & Machine Learning

Pandas / Data Visualization / NumPy

Bioinformatics

Genomic Data Analysis / Immunoinformatics / Computational Biology

Programming

Python / C++ / Perl / C# / MATLAB / PHP

Communication & Collaboration

Advanced video conferencing (Zoom, Teams, Skype, Webex) / project writing / development / management

Soft Skills

Strong analytical and research skills / excellent communication / attentive listener / collaborative team player

CERTIFICATIONS

Software Engineer

1. Software Engineer
2. Machine Learning, AI & Data Science
3. Machine Learning Foundations
4. Python for Data Science
5. Python Programming
6. Virtual Assistant
7. Python (Basic)
8. Python Essentials
9. Python Essentials 2
10. Introduction to AI
11. Problem Solving (BasicS)
12. 3-week course on Free Python Course
13. C++ Programming
14. UI/UX Graphic designer
15. IC3 Digital Literacy Certification
16. Fundamentals of Digital Marketing
17. Building Database Agent
18. JavaScript Programming
19. Java Programming
20. PHP Programming
21. HTML Programming

HONOURS AND AWARDS

[2015]

PEEF Scholarship – Government of Pakistan (2015)

[2019]

EHSAAAS Scholarship – Government of Pakistan (2019)

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s):

English

LISTENING C1 READING C1 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

VOLUNTEERING

Management Head at CFAS, COMSATS University Islamabad

CONFERENCES AND SEMINARS

Seminar on Biosafety and Biosecurity for Researchers/Scientists

REFERENCES

Dr. Muhammad Shoaib

Assistant Professor, COMSATS University Islamabad

Role: MS Thesis Supervisor

Email: mshoaib@cuisahawal.edu.pk

Phone: +92 40 43050013

Dr. Muhammad Asif Rasheed

Assistant Professor, COMSATS University Islamabad

Role: Research Supervisor

Email: asif.rasheed@cuisahawal.edu.pk

Phone: +92 336 301161

Dr. Farhan Akhtar

Assistant Professor, COMSATS University Islamabad

Role: BS Thesis Supervisor

Email: farhan@cuisahawal.edu.pk

Phone: +92 332 4428802